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Title :
OpenMP Program for Matrix Addition Using Threads

Aim of Assignment:
Toimplement matrixaddition using OpenMP threads in C/C++ and understand parallel execution for improving computation speed and efficiency.

H/W and S/W Required:

Hardware Requirements
Intel/AMD Processor
Minimum 4 GB RAM
Keyboard and Monitor

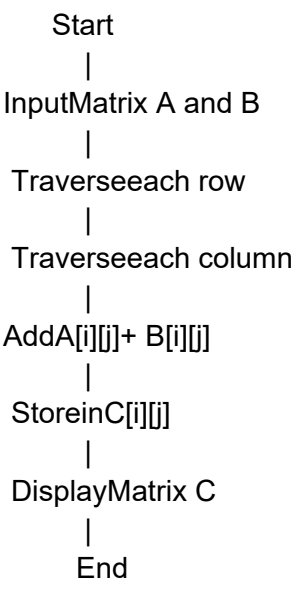
Software Requirements
OperatingSystem: Windows/Linux
GCC Compiler with OpenMP support
Code Editor (VS Code / CodeBlocks / Dev C++)

Problem Statement
Write an OpenMPprogram to perform addition of two matrices using parallel threads. The program should divide the work among multiple threads so that matrix elements are added simultaneously.

Sequential Solution Logic with Diagram

Sequential Logic
1)Start the program.
2)Declare two matrices A and B.
3)Input matrix elements.
4)Traverse matrix using nested loops.
5)Add corresponding elements:
6)Display resultant matrix.
7)Stop.

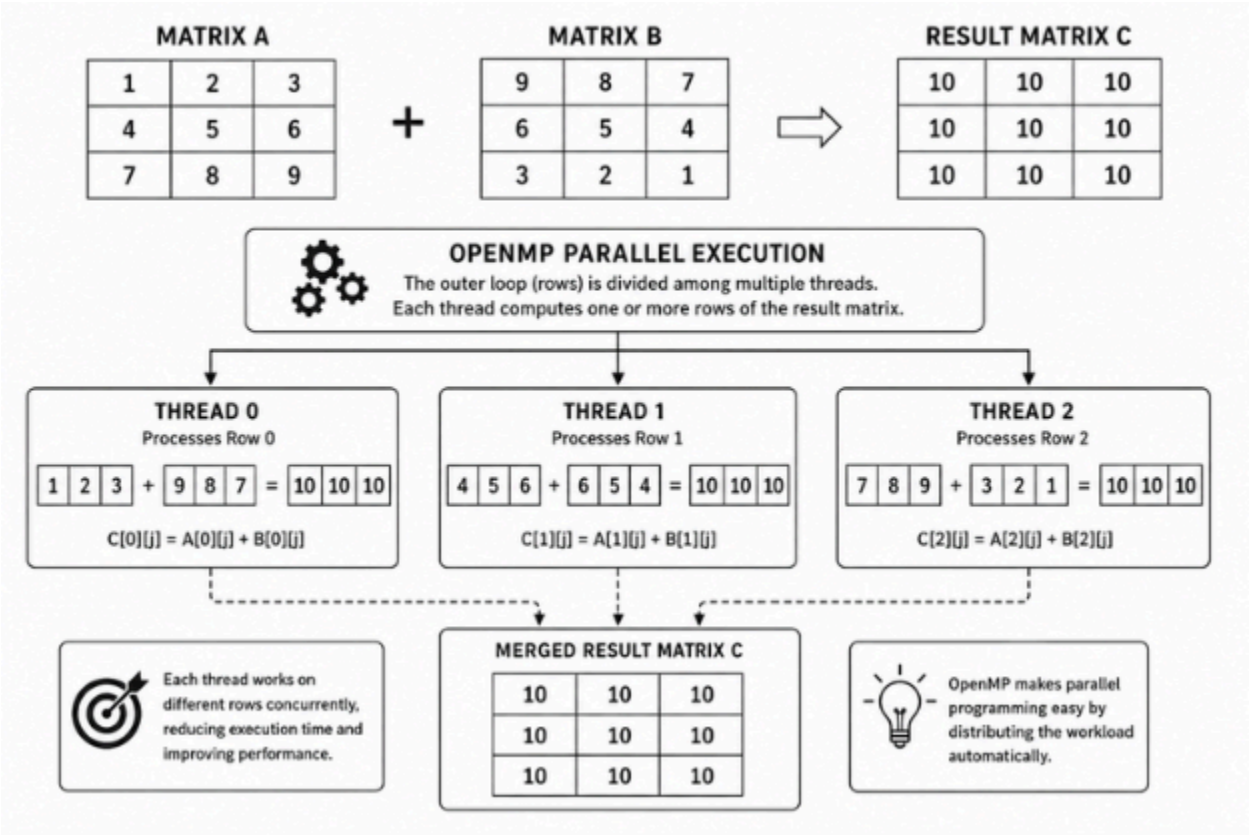
Sequential Flow Diagram



ParallelApproach with Diagram

Parallel Approach
In OpenMP, multiple threads execute matrix addition simultaneously.
1)Different rows/elements are divided among threads.
2)Each thread independently computes part of the resultant matrix.
3)Parallel execution reduces execution time.

OpenMP uses:
#pragma omp parallel for
to distribute loop iterations among threads



Code with Explanation

OpenMP Program for Matrix Additionode with Explanation

```
#include <stdio.h>
#include <omp.h>

int main()
{
   int a[3][3], b[3][3], c[3][3];
   int i, j;

   printf("Enter elements of Matrix A:\n");
   for(i = 0; i < 3; i++)
   {
      for(j = 0; j < 3; j++)
      {
         scanf("%d", &a[i][j]);
```

```
        }
    }

    printf("Enterelements of Matrix B:\n");
    for(i = 0; i < 3; i++)
    {
        for(j=0;j<3;j++)
        {
            scanf("%d",&b[i][j]);
        }
    }

    #pragmaompparallel for private(j)
    for(i = 0; i < 3; i++)
    {
        for(j=0;j<3;j++)
        {
            c[i][j]=a[i][j]+b[i][j];

            printf("Thread%d computed c[%d][%d]\n",
                  omp_get_thread_num(), i, j);
        }
    }

    printf("\nResultantMatrix:\n");

    for(i = 0; i < 3; i++)
    {
        for(j=0;j<3;j++)
        {
            printf("%d",c[i][j]);
        }
        printf("\n");
    }

    return 0;
}
```

Explanation of Program

This program performs Matrix Addition using OpenMP threads on a multicore CPU. Matrix addition is an operation in which corresponding elements of two matrices are added to generate a resultant matrix.

Matrix Addition Formula
C[i][j]=A[i][j]+ B[i][j]
Where:
A[i][j] = element of first matrix
B[i][j] = element of second matrix
C[i][j]= resultant matrix element

The program declares three matrices:
Matrix A stores elements of the first matrix.
Matrix B stores elements of the second matrix.
Matrix C stores the resultant matrix after addition.
The user enters elements of matrices A and B using nested loops. The outer loop traverses rows, while the inner loop traverses columns of the matrices.

Matrix A:
1 2 3
4 5 6
7 8 9

Matrix B:
9 8 7
6 5 4
3 2 1

The addition operation is performed as:

C[0][0] = 1 + 9 = 10
C[0][1] = 2 + 8 = 10
C[0][2] = 3 + 7 = 10

C[1][0] = 4 + 6 = 10
C[1][1] = 5 + 5 = 10
C[1][2] = 6 + 4 = 10

C[2][0] = 7 + 3 = 10
C[2][1] = 8 + 2 = 10
C[2][2] = 9 + 1 = 10

The OpenMP directive:
#pragma omp parallel for

is applied before the outer loop. This directive creates multiple threads and automatically distributes loop iterations among them. Each thread processes different rows of the matrix simultaneously, enabling parallel execution.
For example:
Thread 0 may process Row 0
Thread 1 may process Row 1
Thread 2 may process Row 2
The statement:
c[i][j] = a[i][j] + b[i][j];

adds corresponding elements of matrices A and B and stores the result in matrix C.
The function:
omp_get_thread_num()
returns the ID of the thread currently executing the operation. This helps in observing how multiple threads work concurrently during execution.
After all threads complete execution, the resultant matrix is displayed using nested loops.

Final Output:
10 10 10
10 10 10
10 10 10

The program demonstrates how OpenMP improves computational efficiency by executing matrix operations concurrently instead of sequentially. Parallel execution reduces computation time and utilizes multicore processors effectively, especially for large matrices.

Output :

```c
#include <stdio.h>
#include <omp.h>

int main()
{
    int a[10][10], b[10][10], c[10][10];
    int i, j, r, col;

    printf("Enter number of rows and columns: ");
    scanf("%d%d", &r, &col);

    printf("Enter elements of Matrix A:\n");
    for(i = 0; i < r; i++)
    {
        for(j = 0; j < col; j++)
        {
            scanf("%d", &a[i][j]);
        }
    }

    printf("Enter elements of Matrix B:\n");
    for(i = 0; i < r; i++)
    {
        for(j = 0; j < col; j++)
        {
            scanf("%d", &b[i][j]);
        }
    }

    #pragma omp parallel for private(j)
    for(i = 0; i < r; i++)
    {
        for(j = 0; j < col; j++)
        {
            c[i][j] = a[i][j] + b[i][j];
        }
    }

    printf("Resultant Matrix:\n");
    for(i = 0; i < r; i++)
    {
        for(j = 0; j < col; j++)
        {
            printf("%d ", c[i][j]);
        }
        printf("\n");
    }

    return 0;
}
```

```
ragini@ragini-ThinkPad-E15:~$ gcc -fopenmp matrix_addition.c -o matrix
ragini@ragini-ThinkPad-E15:~$ ./matrix
Enter number of rows and columns:  8 8
Enter elements of Matrix A:
1 2 3 4 5 6 7 8
9 10 11 12 13 14 15 16
17 18 19 20 21 22 23 24
25 26 27 28 29 30 31 32
33 34 35 36 37 38 39 40
41 42 43 44 45 46 47 48
49 50 51 52 53 54 55 56
57 58 59 60 61 62 63 64
Enter elements of Matrix B:
64 63 62 61 60 59 58 57
56 55 54 53 52 51 50 49
48 47 46 45 44 43 42 41
40 39 38 37 36 35 34 33
32 31 30 29 28 27 26 25
24 23 22 21 20 19 18 17
16 15 14 13 12 11 10 9
8 7 6 5 4 3 2 1
Resultant Matrix:
65 65 65 65 65 65 65 65
65 65 65 65 65 65 65 65
65 65 65 65 65 65 65 65
65 65 65 65 65 65 65 65
65 65 65 65 65 65 65 65
65 65 65 65 65 65 65 65
65 65 65 65 65 65 65 65
65 65 65 65 65 65 65 65
ragini@ragini-ThinkPad-E15:~$
```

Conclusion :

The OpenMPprogram for Matrix Addition was successfully implemented using parallel threads. The programadded corresponding elements of two matrices and generated the resultant matrix efficiently. OpenMP enabled parallel execution of matrix operations, improving CPU utilization and reducingexecution time compared to sequential execution.